

RA-ASF

Risk-Aware Adaptive Sensor Fusion for Adverse Weather Autonomous Navigation

Intelligent and Autonomous Systems (IAS)

Team : Tri-Module Architecture

Why Conventional AV Perception Fails

Autonomous vehicles fail catastrophically during sensor degradation — heavy fog, rain, or hardware faults — because conventional systems assign **fixed weights** to sensor inputs, leading to **confident-but-wrong perception**.

Objective 1

Continuously **monitor the health of three sensors** in real-time to detect degradation as it happens.

Objective 2

Dynamically **activate the best available sensor-pair fusion module** based on which sensor is degraded.

Objective 3

Use the active module's **perception uncertainty** to dynamically adjust vehicle safety margins — speed and braking — before a violation occurs.

Relevance to IAS Course: Directly implements Unit 1 (fail-safe mechanisms), Unit 2 (learning-based decision-making under uncertainty), Unit 3 (multi-sensor fusion), and Unit 5 (PID/Stanley motion control) into a single cohesive architecture.

Evolution of Sensor Fusion

01

Early Systems — Static Fusion

Assumed all sensors are always reliable. Failed catastrophically in fog, rain, glare, and hardware faults with no adaptation mechanism.

02

Key Research Progress (2024–2026)

Adaptive fusion (SAMFusion), uncertainty-aware models (UA-Fusion), sensor dropout handling, and CARLA-based fault testing (FADE) emerged as significant advances.

03

Core Limitation — Still Unresolved

Most systems still use **fixed fusion strategies**, lack real-time sensor health awareness, and do **not** adapt vehicle control using perception uncertainty.

What Is Missing — And What We Do Differently

Identified Research Gaps

Gap 1: Static Fusion

Same fusion logic in all conditions; no runtime adaptation to changing sensor quality.

Gap 2: No Sensor Health Awareness

No real-time diagnosis of sensor quality; systems depend on predefined, static conditions.

Gap 3: No Uncertainty → Control Link

Uncertainty stays in perception; vehicle behavior does not adapt based on confidence.

Our Contribution — RA-ASF

→ Real-time Sensor Health Monitoring

Live diagnosis of camera blur, LiDAR point density, and RADAR SNR every frame.

→ Dynamic Module Switching

Three independent fusion pipelines; the best available pair is activated at runtime.

→ Uncertainty-Aware Control

Perception uncertainty feeds directly into a Risk-Aware PID Controller to adjust speed and braking margins.



We close the loop: Perception → Uncertainty → Control

Proposed Approach: Tri-Module Architecture

No single sensor is reliable in all conditions — but **any two sensors together are sufficient**. We utilize three independent pipelines targeting specific weather failure modes.

Module 1

Camera + LiDAR

- **Camera Branch:** YOLOv8n
- **LiDAR Branch:** PointNet++ (lightweight)
- **Fusion:** Feature concatenation + MLP head

Module 2

Camera + RADAR

- **Camera Branch:** YOLOv8n
- **RADAR Branch:** DBSCAN clustering + MLP
- **Fusion:** Hungarian matching + confidence merging

Module 3

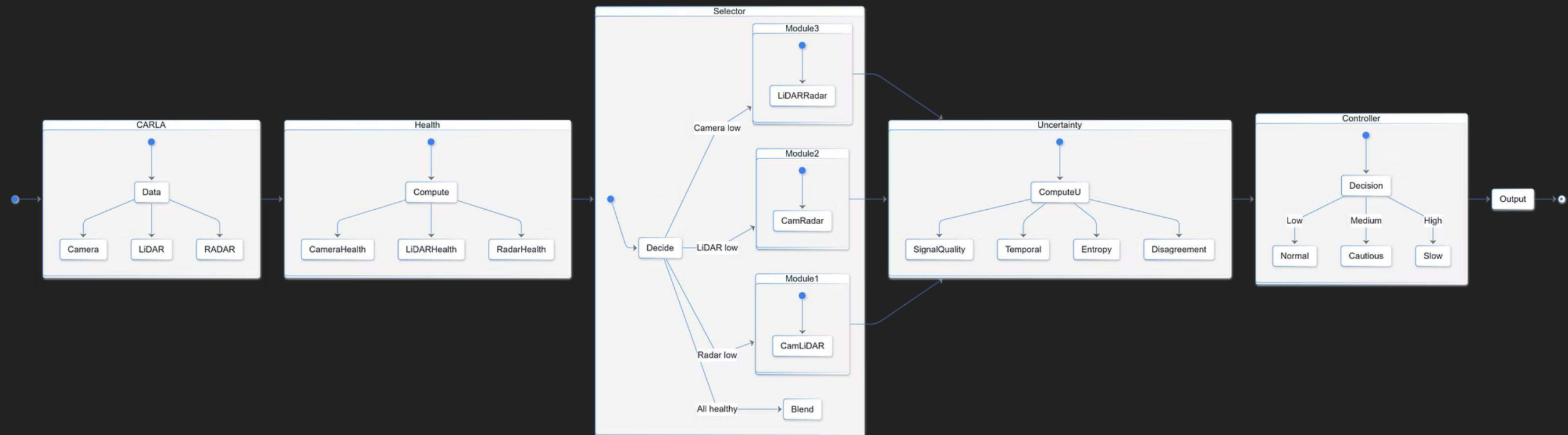
LiDAR + RADAR

- **LiDAR Branch:** PointNet++ (mini)
- **RADAR Branch:** MLP on range-velocity features
- **Fusion:** Feature concatenation + detection head

System Intelligence: A **Fusion Selector** checks live sensor health (blur, point density, SNR) and routes data to the best module. The active module's **uncertainty score** then feeds a **Risk-Aware PID Controller** to dictate vehicle speed and safety margins.

System Block Diagram

Data flows from the CARLA simulation environment through sensor health monitoring and dynamic fusion selection, ultimately commanding the vehicle via a Risk-Aware PID Controller.



Uncertainty Computation : $U_GLOBAL = 1 - (0.5 * Camera_Health + 0.3LiDAR_Healthy + 0.2RADAR_HEALTH)$

(OR)

Uncertainty Computation : $U_GLOBAL = 0.3 * Entropy + 0.2 * Temporal\ Stability + 0.5 * Sensor\ Disagreement$



IMPLEMENTATION

Simulation Progress

Platform: CARLA Simulator (v0.9.14) — selected after Gazebo, Webots, and Python libraries proved insufficient for complex multi-sensor weather simulation needs.

01

CARLA Environment Setup

Ego vehicle spawned in Town03 with fully synchronized Camera, LiDAR, and RADAR sensors attached.

02

Multi-Modal Data Collection

Dataset frames collected across four weather conditions: **CLEAR**, **FOG_LIGHT**, **FOG_HEAVY**, and **RAIN** for all three fusion module training sets.

03

Sensor Health Computation

Logic implemented to compute blur/exposure (camera), point density (LiDAR), and SNR (RADAR) live per frame.

04

Sensor Selection Logic

Core routing logic for degraded sensor detection and module switching is fully functional.

Sensor Health Estimation: Camera, LiDAR, RADAR

Robust, real-time sensor health monitoring provides the critical input for dynamic fusion module selection, ensuring perception integrity under varying conditions.



Camera Health Estimation

Metrics Used:

- **Blur Score (Primary Signal):** Computed using Laplacian variance. Detects fog, motion blur, and defocus.
- **Exposure Score:** Based on deviation from ideal brightness (128). Captures overexposure (glare) and underexposure (darkness).

Final Health: $\text{Health} = 0.65 \times \text{Blur} + 0.35 \times \text{Exposure}$

Note: Blur is prioritized as it is the strongest indicator of environmental degradation.



LiDAR Health Estimation

Metrics Used:

- **Point Count Score:** Ratio of actual returns to expected returns. Strong indicator of fog or signal loss.
- **Intensity Score:** Measures reflection strength.
- **Angular Coverage Score:** Evaluates how much of the 360° scan contains valid returns.

Final Health: $\text{Health} = 0.50 \times \text{Count} + 0.30 \times \text{Intensity} + 0.20 \times \text{Coverage}$

Note: Point count dominates since LiDAR degradation mainly reduces returns.



RADAR Health Estimation

Metrics Used:

- **Detection Count Score:** Number of detected objects vs expected.
- **Velocity Spread Score:** Standard deviation of detected velocities. Ensures diversity and consistency in returns.

Final Health: $\text{Health} = 0.70 \times \text{Count} + 0.30 \times \text{Velocity Spread}$

Special Case: If no detections → health = 0.5 (uncertain state). This prevents false failure detection in empty environments.

Preliminary Results, Challenges & Mitigations

- 📄 ☒ **Initial Observation:** The Sensor Health Monitor successfully computes scores and penalizes simulated sensor dropouts in real-time.

Challenge 1: CARLA Memory Crashes

Rendering dynamic weather requires heavy VRAM, causing hardware crashes during extended simulation sessions.

Solution

Utilized CARLA's `-RenderOffScreen` mode during data collection, reducing VRAM usage by approximately **50%**.

Challenge 2: Sensor Misalignment

Initial LiDAR-to-Camera projection showed spatial misalignment, corrupting early fusion attempts and reducing detection accuracy.

Solution

Fixed by computing exact extrinsic calibration matrices using CARLA's `get_transform()` functions, achieving accurate sensor-space alignment.

Future Contributions — Tri-Module Architecture

Equal contribution across three specialized modules, each member owning an end-to-end pipeline component.



Member 1

Camera + LiDAR Fusion Module

Owens the CARLA simulation pipeline setup, dataset generation, and LiDAR-to-camera projection using YOLOv8-nano.



Member 2

Camera + RADAR Fusion Module

Owens Evidential Deep Learning (EDL) uncertainty quantification and Camera-RADAR Hungarian association logic.



Member 3

LiDAR + RADAR Fusion + Control

Owens the fallback perception module, Sensor Health Monitor, Fusion Selector logic, and the Risk-Aware PID Controller.



Road to Final Review

Next Steps

Step 1 — Train Fusion Modules

Train all three sensor-pair fusion models on the collected adverse weather datasets and calibrate the uncertainty Using Signal Quality, Sensor Disagreement And Entropy.

1

2

Step 2 — Wire Into Live CARLA Loop

Integrate the three independent models into the active Fusion Selector logic inside the live CARLA simulation loop.

3

Step 3 — Run Evaluation Episodes

Conduct 20 full evaluation episodes comparing RA-ASF against an equal-weight baseline system.

4

Step 4 — Generate Final Report Graphs

Produce Collision Rate, Precision-Recall curves, and Uncertainty Calibration plots for the final report submission.